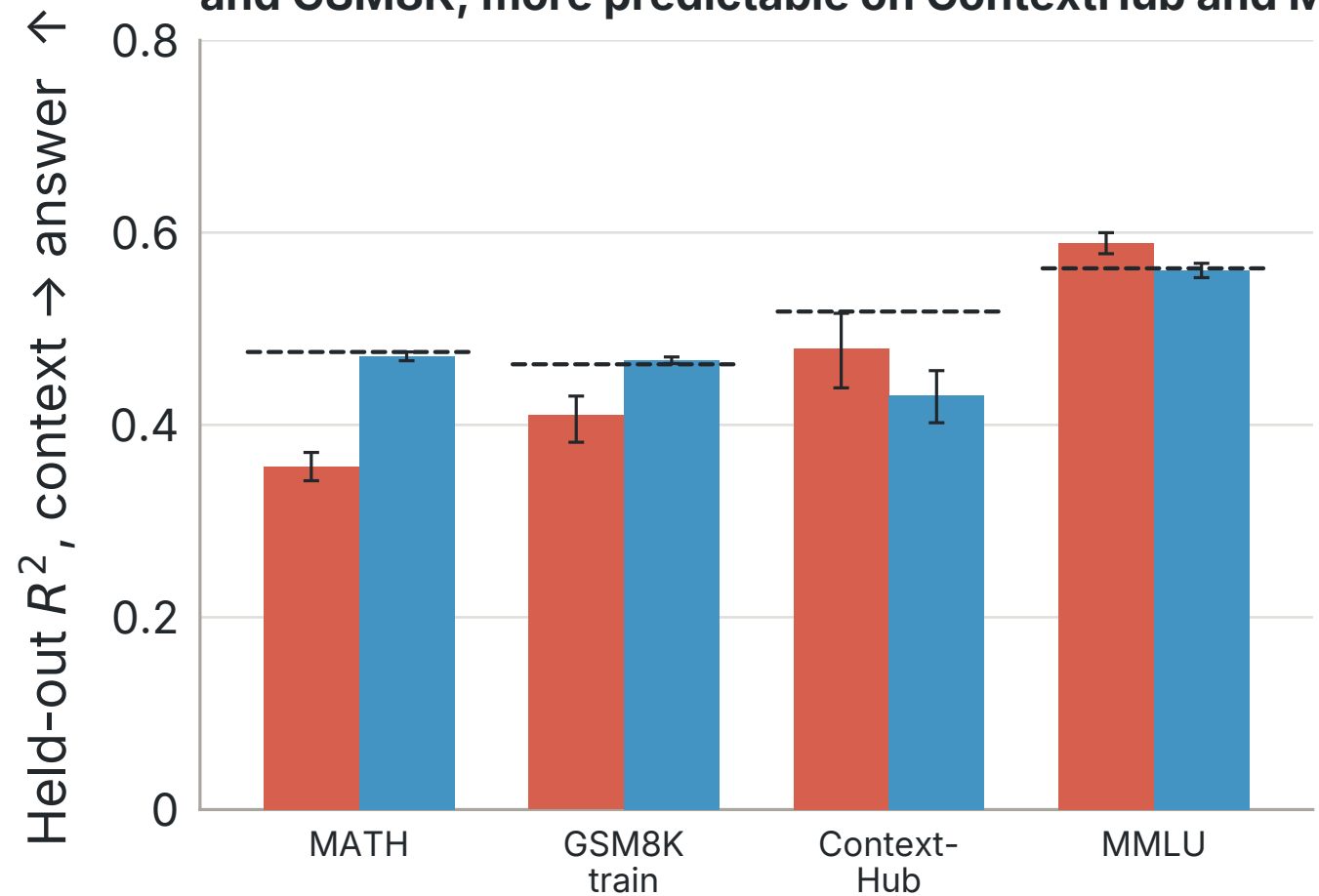


QUESTIONS, LABELED BY WHETHER REASONING WAS NEEDED

Reasoning necessary (right with reasoning, wrong without) Reasoning not needed (right both ways) ---- Whole corpus

A · OPENTHINKER3-7B, VS ITS NON-REASONING PARENT QWEN2.5-7B-INSTRUCT, LAYER 19 · QWEN3-8B, THINKING ON VS THINKING OFF, SAME WEIGHTS, LAYER 24

Necessary questions are less predictable on MATH and GSM8K, more predictable on ContextHub and MMLU



Same on the thinking toggle, except that MMLU is slightly lower

